

Assignment 2

EL2700 Model Predictive Control

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1 Question 1

Implementation only...

2 Question 2

- laptime without curvature optimization: 01:09.415
- laptime with curvature optimization (cost= $\sum_i \|a_i\|_2 \cdot \Delta s$): 00:51.894
- laptime with curvature optimization (cost= $\sum_i \|a_i\|_2 \cdot \Delta s$): 01:07.263

Table 1: Lap time comparison

Scenario	Laptime
Without curvature optimization	01:09.415
With curvature optimization (objective $\sum_i \ a_i\ _2 \cdot \Delta s$)	00:51.894
With curvature optimization (objective $\sum_i \ a_i\ _2 \cdot \Delta s$)	01:07.263

3 Question 3

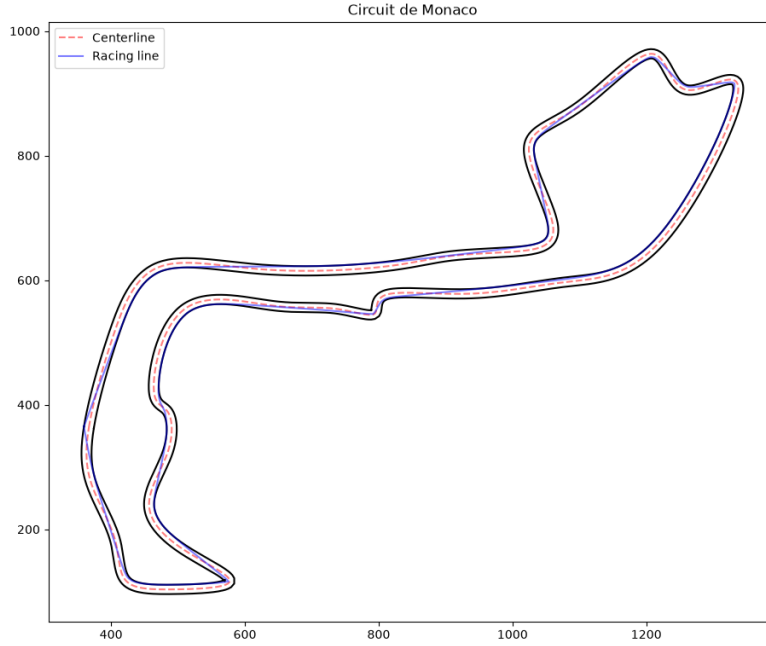


Figure 1: Centerline and Race Line($\text{cost} = \sum_i \|a_i\|_2 \cdot \Delta s$)

With the L1-norm cost, the optimizer suppresses acceleration to very small values at the vast majority of points, yielding large turning radii and high permissible speeds along most segments of the track, so the racing line is significantly faster than the center line. However, the L1 cost permits large acceleration (sharp cuts and kinks in the trajectory) at a few isolated locations. Consequently, on some corners, the curvature of the racing line is a lot higher than that of the centerline. Apparently the total cost is minimized by a raceline that is very smooth on most parts of the track but broken up by hard "cuts" in the racing line.

4 Question 4

4.1 a

The plot for the sum of squares cost function looks a lot more like the expected result. It is shown in figure 2.



Figure 2: Race line plot using $\sum_i \|a_i\|_2^2 \cdot \Delta s$ as cost

The racing line obtained with the squared-norm cost $\sum_i \|a_i\|_2^2 \Delta s$ appears significantly smoother and more realistic than the line from Question 3. It is shown in Figure 2.

While the plot matches the expected result, the laptime is much higher compared to the laptime of the racing line from Question 3. However the laptime calculated in Question 3 will most likely not be realistic, since the turning rate (yaw-rate) of the cars is not fast enough to follow the very sharp turns. Additionally while calculating the laptime in Question 3 the constraints of the longitudinal acceleration of the car are not considered. Before and after a very sharp turn, as they occur in the racing line of Question 3, the car must decelerate and accelerate longitudinally; this acceleration is however also constrained by brakes and wheel torque. These factors are not taken into account while calculating the laptime, thus the result for Question 3 is only a theoretical simulation value and cannot be achieved in practice.

4.2 b

This difference of the two racing lines can be explained by high curvatures receiving a higher penalty for the sum of squares approach. If the curvatures are squared, then high curvatures are penalized harder than low curvatures,

leading to a more even "curvature distribution" over the racing line.